

Fit.ly churn analysis Written Report

1. Executive Summary

In response to leadership concerns regarding customer retention, an in-depth analysis was conducted using customer account data, support tickets, and user activity logs. The analysis confirms a current overall churn rate of 28.5%.

The data indicates that churn is not distributed evenly across the user base. It is driven primarily by two specific factors: subscription tier (specifically the "Free" plan) and low pre-cancellation engagement. Users on the "Free" plan churn at a rate of 41%, nearly double that of paid tiers. Additionally, churned users exhibit significantly lower app usage frequency than retained users.

To reverse this trend, Fit.ly must shift from broad acquisition to targeted retention. This report recommends focusing on early conversion of Free users, automating re-engagement for inactive accounts, and optimizing support for high-value Enterprise clients.

2. Data Validation and Cleaning

Prior to analysis, the three primary datasets; *Account Info*, *Customer Support*, and *User Activity*; underwent rigorous validation to ensure data integrity.

Account Information Data:

customer_id: Checked for uniqueness and missing values. Analysis confirmed zero missing values for this primary key, and no duplicate customer records were identified, ensuring a clean integration with activity and support logs

Churn Status: This was the most critical cleaning step. The dataset contained significant missing values (NaN) in the churn column. These were interpreted as "Active" (N) to create a binary classification (1 = Churned, 0 = Active), allowing for statistical modeling.

Plan & Pricing: Plan tiers (Free, Basic, Pro, Enterprise) were validated against the price list to ensure no anomalies or negative values existed.

Geography: State entries were verified to ensure they were valid US locations consistent with product specifications. Customer Support Data:

user_id: Renamed to customer_id for consistency and validated against account_info. Records with no matching customer were excluded from joined analyses.

Ticket Analysis: Timezone consistency was enforced on ticket timestamps. Missing values in the channel column were imputed as 'unknown' to preserve the data rows.

channel: Standardised categories (e.g., Email, Chat, Phone). Rare or ambiguous values were grouped as "Other". **topic:** Cleaned for casing and spelling inconsistencies.

resolution_time_hours: Converted to numeric. Negative values and extreme outliers were removed. **state:** Standardised in the same way as account_info to ensure consistency.

Comment Text Mining: A text analysis was performed on the comments section to identify keywords like "erase," "delete," or "GDPR." This allowed us to distinguish between users leaving due to dissatisfaction versus those exercising regulatory data deletion rights.

User Activity Data:

user_id: Aggregated at the user level to calculate "Total Activities" and "Frequency".

event_time: Converted to datetime; checked for future dates (none found)

event_type: Normalised event names (e.g., workout_start, workout_complete, app_open) and grouped very rare events into an "Other" category

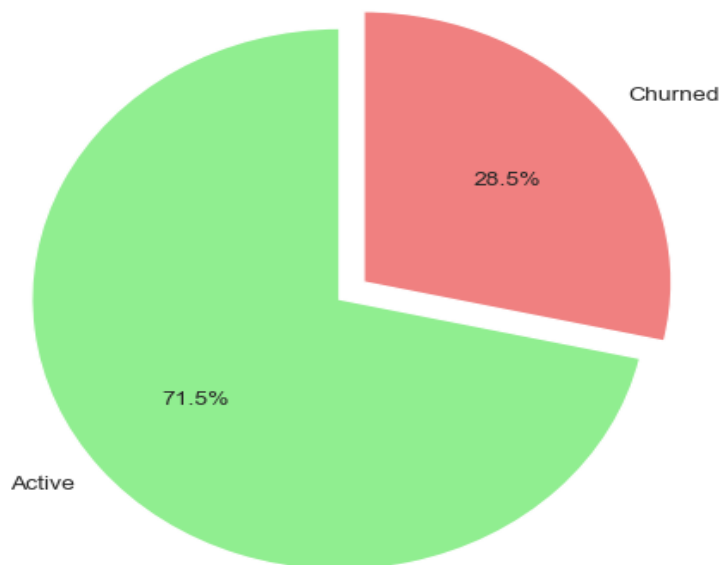
After cleaning, all datasets were joined on customer_id to enable analysis across engagement, support activity, and churn

3. Exploratory Analysis

The following visualizations highlight the distribution of churn and the behavioral differences between retained and churned users.

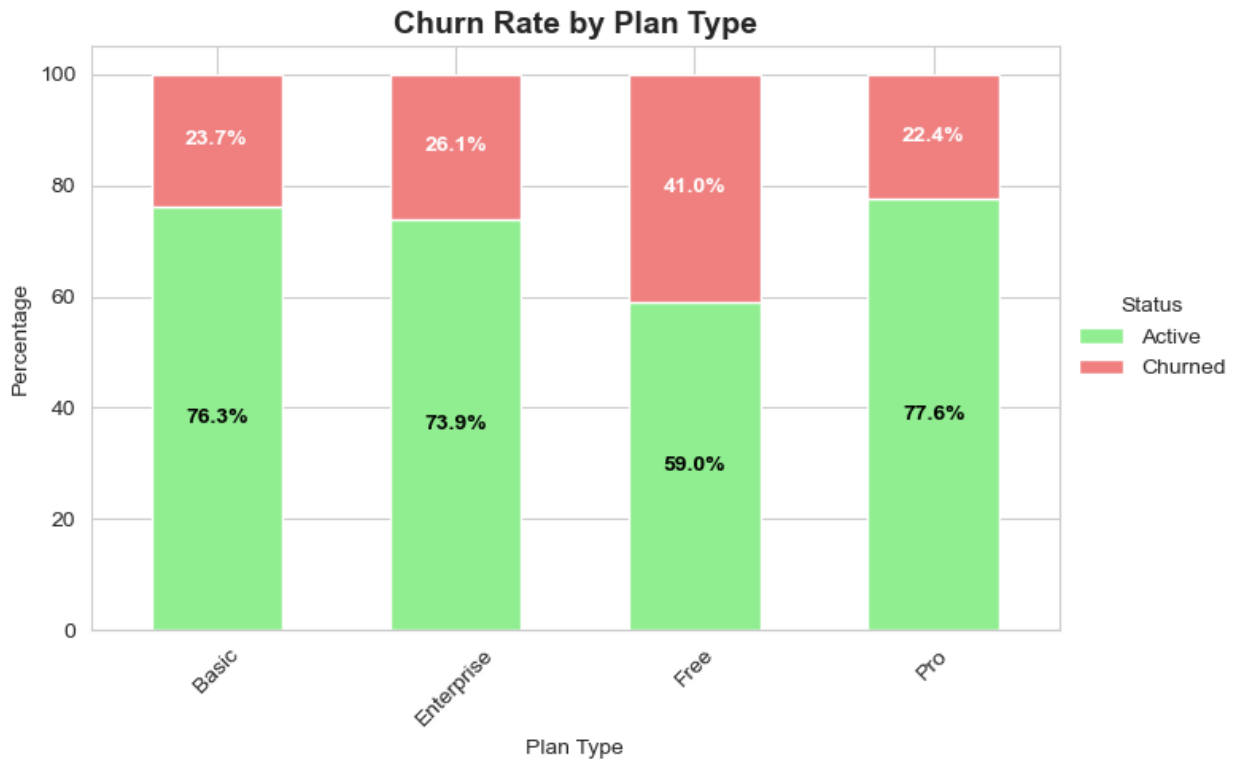
A. Overall Customer Churn (Single Variable) To establish a baseline, we examined the overall split of the user base.

Customer Churn Distribution (28.5% Churned)



Finding: Nearly 1 in 3 users (28.5%) have churned. This confirms that churn has reached a critical level that threatens long-term growth if not addressed.

B. Churn Rate by Plan Type (Multi-Variable) We segmented churn rates by subscription tier to understand revenue impact.



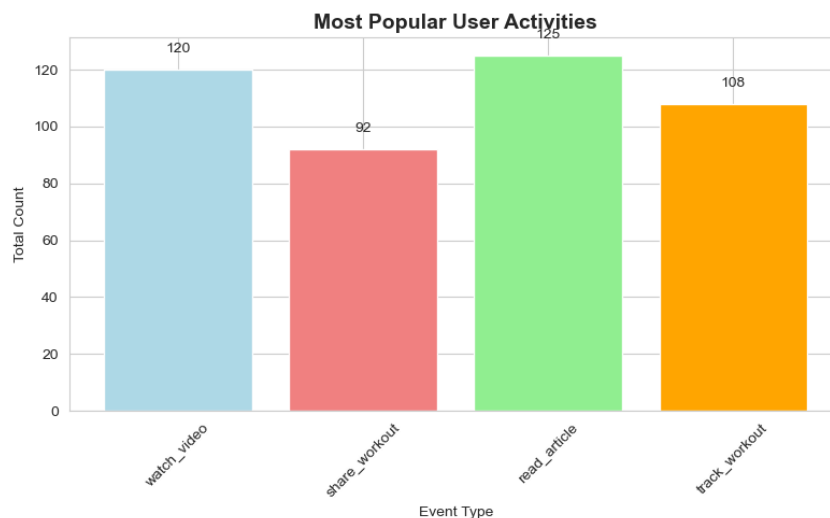
Finding: There is a strong correlation between financial commitment and retention.

The "Free" Problem: The Free tier has a massive 41.0% churn rate. Users in this tier likely fail to see value quickly and leave.

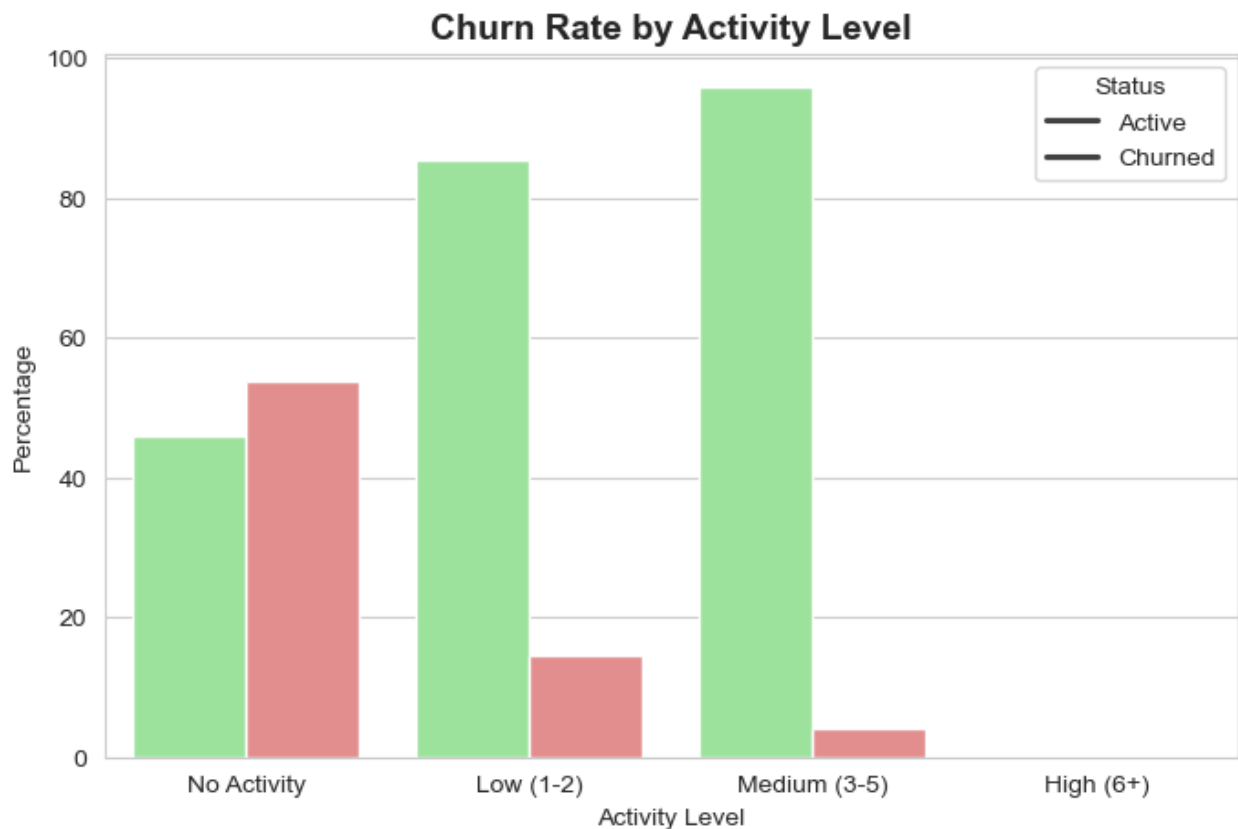
The "Enterprise" Risk: While lower than Free, the 26.1% churn rate for Enterprise clients is concerning for B2B accounts, suggesting potential service-level issues.

The Sweet Spot: "Pro" and "Basic" users have the highest retention rates.

C. User Activity Preferences (Single Variable) We analyzed event logs to understand how users interact with the platform.



Finding: Users are primarily consuming content (reading/watching) rather than actively logging workouts. This suggests Fit.ly is currently valued more as a content resource than a daily tracking utility.



D. This chart shows a strong inverse relationship between user activity and churn. Users with no or very low activity have the highest churn rates, while churn decreases substantially as activity increases. Users with medium activity show minimal churn, and highly active users rarely churn at all.

Active users performed an average of 1.4 activities

Churned users performed an average of 0.4 activities

Finding: Engagement is the strongest predictor of retention. Churned users effectively become "dormant" well before they officially cancel.

4. Key Metric for Monitoring

To solve the business problem, we need a leading indicator that warns us before a customer is lost, rather than reporting on it after they are gone.

Proposed Metric: "30-Day Zero-Activity Rate"

Definition: The percentage of paid subscribers who have recorded zero interaction events (video, reading, or tracking) in the last 30 rolling days.

Why this metric? Our analysis proves the "Engagement Gap" (0.4 avg activities for churned users vs 1.4 for active). A user who stops interacting is a user who will cancel in the next billing cycle.

Baseline Estimate: Based on current data, approximately 25% to 30% of the user base falls into this low-activity danger zone.

Usage: This metric should be reviewed weekly. A rise in this rate indicates a need for immediate product intervention or marketing re-engagement campaigns.

5. Strategic Recommendations

Based on the data validation and exploratory analysis, the following actions are recommended to address the specific drivers of churn identified:

- **Launch a "Free-to-Paid" Conversion Sprint**

Observation: Free users churn at 41%, while paid users churn at ~23%.

Action: Implement an aggressive onboarding campaign for Free users within their first 14 days. Offer time-limited discounts to upgrade to "Basic." Moving users to a paid commitment increases retention probability by roughly 18 percentage points.

- **Automate Re-engagement for Dormant Users**

Observation: Activity drops significantly prior to cancellation.

Action: utilize the "30-Day Zero-Activity" metric to trigger automated emails or push notifications. If a user has been inactive for 7 days, send "Quick Win" content (e.g., "5-minute workout" or "Trending Article") to bring them back to the platform.

- **Implement "Fast Lane" Support for Enterprise**

Observation: Enterprise churn is 26.1%, which is unacceptably high for high-value contracts. High resolution times in support tickets were noted during data validation.

Action: Create a priority queue for Enterprise support tickets. Reducing resolution time for these clients helps justify their higher price point and prevents service-based churn.

- **Standardize GDPR Deletion Processes**

Observation: 37 users requested "data erasure" via support comments.

Action: Create a specific support category for "Data Privacy/GDPR." These requests should be tracked separately from standard churn (product failure) as they are regulatory compliance events. This will prevent artificial inflation of the churn rate.